Mondrian Conformal Prediction for Uncertainty Quantification in ER Discrete-Event Simulation Surrogates

Team : B9

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Course: 23AID201

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Introduction

- Emergency departments are high-pressure queueing systems where staffing decisions directly affect patient wait times.
- Discrete-event simulation is the standard tool for modelling them, but is too slow for large-scale exploration.
- Machine learning surrogates solve the speed problem, at the cost of unclear reliability.
- Uncertainty quantification is the field concerned with attaching confidence to such predictions.
- This project evaluates one UQ method — Conformal Prediction — in the ER simulation setting.

Literature Review

Review is organized around the base paper and three supporting research areas:

Base paper: Gopakumar et al. (2026) validate Conformal Prediction for surrogate-model UQ in physics domains (PDEs, MHD, weather, fusion), and explicitly flag two untested limitations - marginal coverage and the exchangeability assumption.

Conformal Prediction Foundations

Split conformal prediction, coverage guarantees, and category-conditional (Mondrian) variants of CP.

Surrogate Modeling

Learned approximations (gradient boosting, GP, neural nets) for expensive simulations, and their use in decision-making.

Queueing Theory / Discrete-Event Simulation

Discrete-event simulation of service systems, particularly emergency department queueing and staffing models.

Target: 30 papers total, 3 core papers reviewed in depth. In progress.

Research Gap

Gopakumar et al. (2026) validate Conformal Prediction for surrogate-model UQ —

but only in physics domains: PDEs, magnetohydrodynamics (MHD), weather, fusion.

The paper states two explicit limitations:

- Marginal coverage only — no guarantee of coverage within specific subgroups/conditions.
- Relies on an exchangeability assumption between calibration and test data — untested under distribution shift.

Neither limitation has been tested outside physics simulation. Discrete-event / queueing systems are a fundamentally different domain — stochastic discrete arrivals and departures, resource contention, priority scheduling — not continuous PDE fields. This is the gap this project addresses.

Problem Statement

- Surrogate models are fast, learned approximations of expensive simulations, increasingly used to guide decisions in stochastic service systems - ER staffing, queueing networks, etc.
- A point prediction alone isn't enough for a decision like "will adding 2 more doctors bring wait time down enough?" - that needs a calibrated interval, not just a number.
- Gaussian Processes are the established way to get one, but are expensive to train and rely on distributional assumptions rather than a finite-sample guarantee.
- Conformal Prediction (CP) is distribution-free with a finite-sample coverage guarantee and is cheap to calibrate - but it's only been validated for surrogate models in a narrow set of domains so far.

Bridging the Gap: Our Novelty

- Apply Conformal Prediction — standard and Mondrian CP — to a surrogate model of a discrete-event ER queueing simulation, a domain never tested against these limitations before.
  - Marginal coverage limitation: addressed with Mondrian CP, which calibrates separately per category (staffing level x arrival-rate regime) instead of one pooled quantile — targets conditional coverage, not just an average.
  - Exchangeability limitation: addressed with a stress test that pushes the evaluation data outside the training distribution (an out-of-distribution "surge day" scenario) to see where the coverage guarantee actually breaks down.
- Novelty: this is the first empirical test of Conformal Prediction's stated physics-domain limitations in a discrete-event / queueing simulation setting — a direct bridge between the base paper's open questions and a new application domain.

GOAL

Aim & Objectives

Aim: to evaluate whether Conformal Prediction — specifically Mondrian CP — provides reliable, well-calibrated uncertainty quantification for surrogate models of discrete-event simulations, and whether Gopakumar et al.'s stated physics-domain limitations hold in this new domain.

- Build and validate a discrete-event ER simulation calibrated on real hospital arrival data.
- Train a surrogate model to approximate the simulation's outputs.
- Implement a Gaussian Process baseline and standard / Mondrian Conformal Prediction for UQ.
- Test marginal vs. conditional coverage — does Mondrian CP close the gap standard CP leaves open?
- Stress-test the exchangeability assumption under distribution shift.
- Compare all methods on coverage, interval width, and computational cost.

APPROACH

Methodology / Workflow

Discrete-Event Simulation
SimPy, calibrated on real ED data

→

Surrogate Model
Learned approximation of the simulation

→

Uncertainty Quantification
GP baseline vs. standard CP vs. Mondrian CP

Each stage validated against the previous before moving on:

- DES output validated against real aggregated ED statistics before being trusted as a data source.
- Surrogate accuracy validated against DES outputs held out from training.
- Every UQ method evaluated on the same test data and the same target coverage level, so results are directly comparable across methods.
- Dataset Source : Kaggle.com – Used for Validating and Calibrating Discrete Event Simulation

PLAN

Project Roadmap

Phase 1 — Mid-Semester

- Literature review + environment setup
- Explore real hospital dataset, extract calibration distributions
- Build and validate the discrete-event ER simulation
- Generate training data and train the surrogate model
- Implement Gaussian Process baseline for UQ

Phase 2 — End-Semester

- Implement standard Conformal Prediction on surrogate residuals
- Implement Mondrian CP — per-category calibration and coverage
- Stress-test the exchangeability assumption (out-of-distribution scenario)
- Full comparison: GP vs. standard CP vs. Mondrian CP
- Final report and end-semester presentation